OPTIMAL SIZING OF LEVERAGED CRYPTO LONG–SHORT POSITIONS UNDER KOU JUMP-DIFFUSION VIA KILLED MOMENTS

We study a leveraged crypto long–short position, held on a DeFi lending protocol, whose solvency is governed by a collateral-to-liability health factor and whose payoff is killed when that factor crosses a liquidation boundary. Sizing such a position requires the survival probability and killed-payoff moments at every candidate initial buffer. We derive a semi-analytic Laplace-domain method for these quantities under bivariate Kou double-exponential jump–diffusion dynamics. The resulting finite-dimensional system is evaluated by numerical Laplace inversion, avoiding path-sampling noise and making repeated evaluation across candidate buffers practical. [Discussion of the sizing analysis]

Keywords: DeFi lending; crypto assets; leveraged long–short; killed payoff moments; health factor; Kou double-exponential jump diffusion; first-passage problem; Laplace resolvent

JEL Classification: G11; G12; C65

1. Introduction

Crypto-asset DeFi lending protocols such as AAVE and Compound enable a leveraged long–short strategy: an investor deposits one crypto asset as collateral, borrows a second crypto asset, and sells it short. The position is governed by a real-time *health factor*—a borrowing-power-adjusted ratio of collateral to liability—and may be liquidated when that ratio falls below one. The resulting exposure combines leveraged relative-value risk, a path-dependent liquidation boundary, discontinuous price jumps, and state-dependent survival. Its initial health buffer controls both leverage and distance to liquidation, but evaluating that trade-off first requires the distribution of a payoff killed at the liquidation boundary.

The two recurring objects are the probability that the position survives to the horizon and the moments of its killed terminal payoff, and both must be evaluated repeatedly across candidate initial health buffers. Monte Carlo simulation can approximate them, but barrier discretisation and sampling noise are especially consequential near liquidation, and repeated simulation is costly inside a sizing search. The missing input for solving the sizing problem is therefore not a universal economic objective, but a fast, repeatable map from the initial buffer to survival probabilities and killed-payoff moments.

We work under a bivariate Kou double-exponential jump–diffusion model (Kou, 2002) and make four contributions. First, a unit-equity normalization collapses the two-asset holding to a single initial health buffer, the natural sizing parameter, while liquidation depends only on a one-dimensional log-price ratio between the two legs. Second, for each admissible moment order, an exponential change of measure absorbs the short-leg price factor, reducing the two-asset killed moment to a one-dimensional killed expectation of that log-price ratio. Third, the double-exponential jump structure turns the resulting Laplace-domain equation into algebra: in the generic case, three barrier conditions pin down a fixed, low-dimensional linear system whose size does not grow with the moment order; numerical Laplace inversion then recovers the time-domain quantities (Talbot, 1979). Fourth, these objective-independent outputs turn initial-buffer sizing into a one-dimensional optimization, which we illustrate under alternative liquidation constraints and moment-based objectives without taking a position on which objective is economically correct.

This paper sits at the intersection of the DeFi liquidation literature and the analytical first-passage literature for jump diffusions. Qin et al. (2021) and Heimbach and Huang (2024) document the incentive structure and leverage amplification of DeFi liquidations, but do not provide the repeated killed-moment evaluation developed here. Kou and Wang (2003) derive explicit Laplace-transform first-passage results under double-exponential jump diffusion, and Sepp (2004) extends this to barrier option pricing; we instead price a barrier-killed long–short ratio, bridging the two settings via the measure-change reduction above. More distantly, Jacobs et al. (1999) advocate joint optimisation of long and short legs, and Dang and Forsyth (2016) study dynamic mean–variance control under jump risk; our sizing problem is a static analogue, optimizing a single initial buffer against a killed-payoff distribution rather than a control process.

The remainder of the paper is organised as follows. Section 2 defines the position, the liquidation barrier, and the health-buffer parameterization. Section 3 specifies the bivariate Kou model and its tilted dynamics. Section 4 reduces payoff moments to one-dimensional killed expectations, and Section 5 derives their time-Laplace transforms and finite-dimensional evaluation system. Section 6 illustrates the framework with an empirical calibration and health-buffer evaluation map, and Section 7 applies that map to initial-buffer sizing rules. Section 8 concludes. The accompanying *Residual-Shape Calibration and Empirical Drift Construction: Companion Note* describes the calibration estimator and drift construction; Appendix A derives the downward-phase boundary conditions, and Appendix B collects the proofs.

2. Leveraged Long–Short Position and Health-Factor Barrier

We consider a leveraged long–short portfolio in which an investor holds asset S_1 and owes asset S_2, with net initial equity normalized to one unit of account. The position is liquidated if collateral becomes too small relative to the liability. This section defines the barrier, the killed-payoff object studied in the paper, and the mapping from the initial health buffer to holdings. No investor objective is needed for these definitions.

2.1. Health factor, liquidation time, and killed payoff

Let $(\Omega, \mathscr{F}, (\mathscr{F}_t)_{t\geq 0}, \mathbb{P})$ be a filtered probability space. Two strictly positive asset prices S_1, S_2 evolve as

$$S_{i,t} = S_{i,0}e^{X_{i,t}}, \qquad X_{i,0} = 0, \quad i = 1, 2,$$

where X_1, X_2 are log-price processes to be specified in Section 3. The investor holds $n_1 > 0$ units of asset S_1 (long leg) and $n_2 > 0$ units of asset S_2 (short leg).

Let

$$V_{1,t} := n_1 S_{1,t}, \qquad V_{2,t} := n_2 S_{2,t}$$

denote the *collateral value* (long leg) and the *borrow value* (outstanding liability) at time t. Fix $b \in (0,1)$, the *liquidation threshold* of asset S_1 (the Aave v3 parameter of the same name; distinct from the maximum LTV at origination, which is a separate and typically lower protocol constant). Solvency requires $V_{2,t} \leq bV_{1,t}$; liquidation occurs when the borrow value exceeds adjusted collateral,

$$V_{2,t} > bV_{1,t}.$$

The *health factor* summarises this margin condition as a single ratio,

$$H_t := \frac{bV_{1,t}}{V_{2,t}} = \frac{bn_1S_{1,t}}{n_2S_{2,t}}, \qquad h_t := \log H_t, \tag{1}$$

where h_t is the *log health factor*. The solvency condition is $H_t \geq 1$ ($h_t \geq 0$); breaching it is equivalent to the borrow value exceeding the b-adjusted collateral. Liquidation is triggered at the first such instant:

$$\tau := \inf\{t \geq 0 : \ H_t < 1\} = \inf\{t \geq 0 : \ h_t < 0\}.$$

A larger b permits a larger borrow-to-collateral ratio $V_2/V_1 \leq b$ before liquidation, giving the position more room to absorb adverse price moves; a smaller b is a stricter protocol threshold, triggering earlier forced exit and limiting bad-debt exposure.

Since

$$h_t = \log\Big(\frac{bn_1S_{1,0}}{n_2S_{2,0}}\Big) + (X_{1,t} - X_{2,t}),$$

define the initial log health factor and the log-ratio process by

$$h_0 := \log\Big(\frac{bn_1S_{1,0}}{n_2S_{2,0}}\Big), \qquad Z_t := X_{1,t} - X_{2,t},$$

so that $h_t = h_0 + Z_t$ and the liquidation time is equivalently

$$\tau = \inf\{t \geq 0 : \ Z_t < -h_0\}.$$

The one-dimensional process Z thus carries all barrier-relevant risk: the position survives if and only if Z stays above the absorbing level $-h_0$. The parameter $h_0 = \log H_0$ is therefore the *initial log-cushion* — the log-distance of the initial health factor from the liquidation boundary $H = 1$. A larger h_0 means the position starts further from forced exit; $h_0 \to 0^+$ means the position starts almost at the boundary with negligible cushion.

The *killed terminal payoff* at horizon T is

$$\Pi_T := (n_1S_{1,T} - n_2S_{2,T})\,\mathbf{1}_{\{\tau > T\}}.$$

On $\{\tau > T\}$ the health factor satisfies $H_T \geq 1$, so $bn_1S_{1,T} \geq n_2S_{2,T}$; since $b \in (0,1)$ this forces $n_1S_{1,T} > n_2S_{2,T}$, meaning the position carries strictly positive terminal equity on every surviving path. No positive-part truncation is needed. The indicator $\mathbf{1}_{\{\tau>T\}}$ assigns zero to liquidated paths. This defines a *killed payoff*; it is a mathematical convention rather than a model of the investor's protocol-specific recovery. Since the initial net book value is normalised to one unit of account (Section 2.2), net profit on a surviving path is $R_T := \Pi_T - 1$, but all moments in this paper are stated in terms of Π_T directly. This paper characterises the killed moments of Π_T, the conditional distribution on $\{\tau > T\}$, and the probability of that event, with $H_t = 1$ treated as an absorbing boundary. Liquidation mechanics are outside the model scope.

Remark 1 (Recovery). When the boundary is crossed, a lending protocol initiates a protocol-specific, state-dependent liquidation procedure. Partial debt repayment, liquidation bonuses, keeper timing, market liquidity, and execution prices can all affect the value ultimately retained by the borrower (Qin et al., 2021; Heimbach and Huang, 2024). A single-barrier first-passage model does not determine those cash flows.

We therefore set terminal wealth to zero following liquidation solely to define the killed-payoff benchmark Π_T. This convention is deliberately severe and analytically convenient; it is not intended to reproduce recovery cash flows or to assert that recovery has zero conditional mean. If R_τ denotes a specified recovery value carried to T, the corresponding investor-wealth object would instead be

$$W_T = (n_1S_{1,T} - n_2S_{2,T})\mathbf{1}_{\{\tau>T\}} + R_\tau\mathbf{1}_{\{\tau\leq T\}} = \Pi_T + R_\tau\mathbf{1}_{\{\tau\leq T\}}.$$

The closed-form construction in this paper concerns the first term. Modelling the second requires additional protocol and execution assumptions.

Remark 2 (Dynamic collateral management). A practitioner who monitors the health factor could top up collateral or partially unwind the short leg to steer H_t away from the liquidation boundary. Such a strategy would replace the absorbing barrier with a reflecting or controlled boundary and the static evaluation problem with a stochastic control problem. We hold the position fixed to isolate the ex-ante killed-moment calculation; the assumption should not be interpreted as a claim that dynamic intervention is infeasible or suboptimal.

Thus killed payoff and investor terminal wealth are distinct objects. The former is the tractable object evaluated below; recovery and intervention overlays may change the latter and, consequently, any sizing decision built from it.

Using the definition of H_T, one obtains the following factorization.

Lemma 2.1 (Payoff factorization)**.**

$$\Pi_T = \frac{n_2}{b} S_{2,0} e^{X_{2,T}} \left(e^{h_0+Z_T} - b\right) \mathbf{1}_{\{\tau > T\}}. \tag{2}$$

The factor $(e^{h_0+Z_T} - b)$ *is strictly positive on* $\{\tau > T\}$ *because* $H_T = e^{h_0+Z_T} \geq 1 > b$*, consistent with the definition of* Π_T.

This separates the payoff into a numeraire factor driven by S_2, and a barrier-killed option on the log-ratio Z with moneyness controlled by h_0.

2.2. Health-buffer parameterization and feasible set

The two coin holdings (n_1, n_2) represent two degrees of freedom. A normalization reduces this to a single scalar, and the barrier structure identifies h_0 as the natural evaluation parameter. This reduction does not itself select a portfolio.

We impose the unit-initial-value constraint

$$n_1 S_{1,0} - n_2 S_{2,0} = 1,$$

which fixes the net book value at inception. Setting $x := n_2 S_{2,0}$, one has $n_1 S_{1,0} = 1 + x$ and $e^{h_0} = b(1+x)/x$. Solving for x,

$$x = \frac{b}{e^{h_0} - b},$$

which yields the holding map

$$n_2(h_0) = \frac{b}{S_{2,0}(e^{h_0} - b)}, \qquad n_1(h_0) = \frac{e^{h_0}}{S_{1,0}(e^{h_0} - b)}. \tag{3}$$

Positivity of both holdings requires $e^{h_0} > b$, i.e.

$$h_0 > \ln b.$$

This algebraic condition is necessary but not sufficient for a feasible DeFi position. Initial solvency additionally requires $H_0 = e^{h_0} > 1$, hence $h_0 > 0$. If the protocol imposes a maximum loan-to-value ratio $\ell_{\max} \in (0, b)$ at origination, then

$$\frac{V_{2,0}}{V_{1,0}} = \frac{b}{H_0} \leq \ell_{\max} \quad \Longleftrightarrow \quad h_0 \geq h_0^{\min} := \log\left(\frac{b}{\ell_{\max}}\right) > 0.$$

Accordingly, define the feasible set

$$\mathscr{H}(b,\ell_{\max}) := \begin{cases} [\log(b/\ell_{\max}),\infty), & \text{when an origination LTV } \ell_{\max}\in(0,b) \text{ applies,} \\ (0,\infty), & \text{otherwise.} \end{cases} \tag{4}$$

The lower endpoint is included in the first case because the protocol permits origination at its maximum LTV. The point $h_0 = 0$ is excluded in the second case because it starts the position at the liquidation boundary. The initial gross leverage — long-leg value per unit of equity — follows directly from (3):

$$L_0 := n_1(h_0)\, S_{1,0} = \frac{e^{h_0}}{e^{h_0}-b}. \tag{5}$$

L_0 is strictly decreasing in h_0. Algebraically, $L_0 \uparrow \infty$ as $h_0 \downarrow \ln b$, but this limit is infeasible because $\ln b < 0$ and the position would already be liquidatable at inception. Over the feasible set (4), maximum leverage is attained at the smallest permitted health buffer; as $h_0 \to \infty$, leverage $L_0 \downarrow 1$ (the short leg vanishes and the position becomes unlevered). Equivalently, the borrow-to-equity ratio is $L_0 - 1 = b/(e^{h_0}-b)$, which also collapses the initial health factor back to $H_0 = e^{h_0}$ via $H_0 = bL_0/(L_0-1)$. Choosing h_0 is therefore equivalent to choosing the initial leverage.

For every feasible h_0, the objective-independent model output is the evaluation map

$$h_0 \longmapsto \left(p_{\mathrm{liq}}(T,h_0), M_1(T,h_0), \ldots, M_K(T,h_0), \frac{M_1(T,h_0)}{p_{\mathrm{surv}}(T,h_0)}, \ldots, \frac{M_K(T,h_0)}{p_{\mathrm{surv}}(T,h_0)} \right), \tag{6}$$

where $M_k(T,h_0) = \mathbb{E}[\Pi_T^k]$ and K is any fixed admissible integer order. Sections 4–5 construct this map. A decision maker may subsequently apply a liquidation limit or utility criterion to it; alternative examples are kept separate in Section 7.

3. Bivariate Kou Dynamics and Exponential Tilting

3.1. Bivariate Kou model

We assume under $\mathbb{P}$ that each log-price satisfies the SDE

$$\mathrm{d}X_{i,t} = \mu_i^X\,\mathrm{d}t + \sigma_i\,\mathrm{d}B_{i,t} + \mathrm{d}\left(\sum_{n=1}^{N_{i,t}} J_i^{(n)}\right), \qquad i=1,2, \tag{7}$$

where B_1, B_2 are correlated standard Brownian motions with $\mathrm{d}\langle B_1,B_2\rangle_t = \rho\,\mathrm{d}t$, N_i is a Poisson process with intensity λ_i independent of (B_1,B_2), and the jump sizes $J_i^{(n)}$ are i.i.d. with the Kou double-exponential law. The jump density, characteristic function, and moment generating function are

$$f_i(j) = \frac{p_i}{\eta_{i,+}} e^{-j/\eta_{i,+}} \mathbf{1}_{\{j>0\}} + \frac{1-p_i}{\eta_{i,-}} e^{j/\eta_{i,-}} \mathbf{1}_{\{j<0\}}, \qquad \varphi_{J_i}(u) = \frac{p_i}{1-iu\eta_{i,+}} + \frac{1-p_i}{1+iu\eta_{i,-}}, \tag{8}$$

$$M_i(s) = \frac{p_i}{1-s\eta_{i,+}} + \frac{1-p_i}{1+s\eta_{i,-}}, \qquad s \in \left(-\tfrac{1}{\eta_{i,-}}, \tfrac{1}{\eta_{i,+}}\right). \tag{9}$$

Here $\eta_{i,+}$ and $\eta_{i,-}$ are the *means* of the positive and negative jump sizes (not exponential rates); the corresponding rates are $1/\eta_{i,+}$ and $1/\eta_{i,-}$. The constraint $\eta_{i,+} < 1$ is required for the price-jump compensator χ_i defined below to be finite.

For each asset, we use a single free drift parameter, the *expected-price growth exponent* μ_i, defined by

$$\mathbb{E}_{\mathbb{P}}[S_{i,t}/S_{i,0}] = e^{\mu_i t}. \tag{10}$$

This is not the mean log-return rate. Define the price-jump compensator

$$\chi_i := M_i(1) - 1 = \mathbb{E}[e^{J_i} - 1] = \frac{p_i}{1-\eta_{i,+}} + \frac{1-p_i}{1+\eta_{i,-}} - 1. \tag{11}$$

Since $\mathbb{E}_{\mathbb{P}}[e^{X_{i,t}}] = \exp(t[\mu_i^X + \frac{1}{2}\sigma_i^2 + \lambda_i\chi_i])$, matching (10) gives the drift coefficient in the log-price SDE,

$$\mu_i^X := \mu_i - \tfrac{1}{2}\sigma_i^2 - \lambda_i\chi_i. \tag{12}$$

Thus μ_i^X is a derived shorthand used in the characteristic exponent and subsequent calculations, not an additional model parameter.

The bivariate process (X_1, X_2) is a Lévy process with joint characteristic exponent

$$\begin{aligned}\Psi(u,v) := i(\mu_1^X u + \mu_2^X v) - \tfrac{1}{2}\Big(\sigma_1^2 u^2 + \sigma_2^2 v^2 + 2\rho\sigma_1\sigma_2 uv\Big)\\ + \lambda_1\big(\varphi_{J_1}(u) - 1\big) + \lambda_2\big(\varphi_{J_2}(v) - 1\big),\end{aligned} \tag{13}$$

so that $\mathbb{E}_{\mathbb{P}}[e^{i(uX_{1,t}+vX_{2,t})}] = e^{t\Psi(u,v)}$.

3.2. Tilted dynamics

For each integer $k \geq 0$, define the k-tilted measure $\mathbb{P}^{(k)}$ by

$$\left.\frac{\mathrm{d}\mathbb{P}^{(k)}}{\mathrm{d}\mathbb{P}}\right|_{\mathscr{F}_t} = \frac{e^{kX_{2,t}}}{\mathbb{E}_{\mathbb{P}}[e^{kX_{2,t}}]} = \exp\left(kX_{2,t} - t\Psi(0,-ik)\right), \qquad k\eta_{2,+} < 1.$$

Lemma 3.1 (Lévy exponent under $\mathbb{P}^{(k)}$)**.** *Under $\mathbb{P}^{(k)}$, the log-ratio $Z = X_1 - X_2$ is again a Lévy process with exponent*

$$\psi_Z^{(k)}(s) = \mu_Z^{(k)} s + \tfrac{1}{2}\sigma_Z^2 s^2 + \lambda_1\big(M_1(s) - 1\big) + \lambda_2\big(M_2(k-s) - M_2(k)\big), \tag{14}$$

where

$$\sigma_Z^2 := \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2 \geq 0, \qquad \mu_Z^{(k)} := (\mu_1^X - \mu_2^X) - k(\sigma_2^2 - \rho\sigma_1\sigma_2).$$

In particular, the generator satisfies

$$\mathscr{L}^{(k)} e^{sz} = \psi_Z^{(k)}(s)\, e^{sz}. \tag{15}$$

▷ *Proof, p. 26*

This eigenrelation is the key identity used in Section 5 to turn the Laplace-domain problem into a finite exponential representation.

Remark 3 (Superscript (k) convention). A superscript (k) indicates evaluation under $\mathbb{P}^{(k)}$. The untilted case $k = 0$ recovers the original dynamics of Z under $\mathbb{P}$. For $k \geq 1$, the k-th moment $\mathbb{E}[\Pi_T^k]$ is computed under $\mathbb{P}^{(k)}$. The constraint $k\eta_{2,+} < 1$ makes the tilted measure well-defined, while the payoff forcing additionally requires $k\eta_{1,+} < 1$ so that the long-leg exponential moments $M_1(j)$ are finite for $j = 0, \ldots, k$. Thus a payoff-moment order is *admissible* when

$$k\eta_{1,+} < 1, \qquad k\eta_{2,+} < 1. \tag{16}$$

Throughout, k is an integer. There is no restriction to $k = 1$ or $k = 2$: every fixed positive integer satisfying (16) is covered.

For the Kou structure, the tilt transforms the jump rates of Z into the k-dependent phase rates

$$r_{1,+}^{(k)} := \frac{1}{\eta_{1,+}}, \quad r_{1,-}^{(k)} := \frac{1}{\eta_{1,-}}, \quad r_{2,+}^{(k)} := \frac{1}{\eta_{2,-}} + k, \quad r_{2,-}^{(k)} := \frac{1}{\eta_{2,+}} - k.$$

Because $Z = X_1 - X_2$, positive jumps of X_2 appear as downward jumps of Z: the tilt by e^{kX_2} changes the downward phase rate from $1/\eta_{2,+}$ to $1/\eta_{2,+} - k$, which is positive precisely when $k\eta_{2,+} < 1$. The rates $r_{1,-}^{(k)}$ and $r_{2,-}^{(k)}$ enter the finite-dimensional system in Section 5. The associated phase operators and their truncation at the liquidation boundary are developed separately in Appendix A.

Table 1 collects the quantities introduced in this subsection.

Symbol	Definition	Role
$\mathbb{P}^{(k)}$	$\mathrm{d}\mathbb{P}^{(k)}/\mathrm{d}\mathbb{P} \propto e^{kX_{2,t}}$	Measure under which moment reduction is carried out
k	Non-negative integer; (16) for $k \geq 1$	Tilting order; $k \geq 1$ for moments, $k = 0$ for survival probability
$\psi_Z^{(k)}(s)$	Lévy exponent of Z under $\mathbb{P}^{(k)}$	Characteristic root equation $\psi_Z^{(k)}(\gamma) = q$ governs the resolvent
σ_Z^2	$\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$	Variance of the Brownian part of Z; appears in $\psi_Z^{(k)}$
$\mu_Z^{(k)}$	$(\mu_1^X - \mu_2^X) - k(\sigma_2^2 - \rho\sigma_1\sigma_2)$	Drift of Z under $\mathbb{P}^{(k)}$; appears in $\psi_Z^{(k)}$
$\mathscr{L}^{(k)}$	Generator of Z under $\mathbb{P}^{(k)}$	$\mathscr{L}^{(k)} e^{sz} = \psi_Z^{(k)}(s) e^{sz}$; killing is imposed by zero extension in Section 5
$r_{j,\pm}^{(k)}$	Phase rates (see above)	Exponential rates of the jump components of Z under $\mathbb{P}^{(k)}$

Table 1: Tilted-dynamics quantities. Subscript $j = 1$ refers to the contribution from X_1 jumps; $j = 2$ to that from X_2 jumps (sign-reversed because $Z = X_1 - X_2$).

4. Reduction of Killed Payoff Moments

Fix any integer $k \geq 1$ satisfying the two moment-domain conditions (16). Computing $\mathbb{E}_{\mathbb{P}}[\Pi_T^k]$ directly under $\mathbb{P}$ is intractable: the payoff couples the joint process (X_1, X_2) to the barrier event $\{\tau > T\}$ in a way that is neither separable nor analytically tractable. An exponential change of measure resolves this by tilting the path measure by $e^{kX_{2,T}}$, which absorbs the short-leg factor into the density and leaves a purely one-dimensional killed expectation for the log-ratio $Z = X_1 - X_2$ under the k-tilted measure $\mathbb{P}^{(k)}$. Since Z remains a Lévy process under $\mathbb{P}^{(k)}$, no information about the joint dynamics is lost: the tilted characteristic exponent encodes the full marginal law of Z. The k-th moment then factorises as the known scalar $e^{T\Psi(0,-ik)} = \mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}}]$ times the barrier-killed expectation $m_k(T; h_0)$ under $\mathbb{P}^{(k)}$, which is computed via the Laplace–resolvent of Section 5.

Define

$$g_k(z; h_0) := (e^{h_0+z} - b)^k.$$

Since the survival event $\{\tau > T\}$ implies $Z_T > -h_0$, we have $e^{h_0+Z_T} > 1 > b$ on $\{\tau > T\}$, so $g_k(Z_T; h_0) = (e^{h_0+Z_T} - b)^k > 0$ and is a polynomial in e^{Z_T}. From (2),

$$\Pi_T^k = \left(\frac{n_2}{b}\right)^k S_{2,0}^k e^{kX_{2,T}}\, g_k(Z_T; h_0)\, \mathbf{1}_{\{\tau > T\}}.$$

$$\mathbb{E}_{\mathbb{P}}[\Pi_T^k] = \left(\frac{n_2}{b}\right)^k S_{2,0}^k\, e^{T\Psi(0,-ik)}\, \mathbb{E}_{\mathbb{P}^{(k)}}\Big[g_k(Z_T; h_0)\, \mathbf{1}_{\{\tau > T\}}\Big]. \tag{17}$$

Proposition 4.1 (Every admissible integer-order moment)**.** *Let $k \geq 1$ be an integer satisfying* (16)*. Define*

$$m_k(T;h_0) := \mathbb{E}_{\mathbb{P}^{(k)}}\Big[g_k(Z_T;h_0)\,\mathbf{1}_{\{\tau>T\}}\Big].$$

Then

$$\mathbb{E}_{\mathbb{P}}[\Pi_T^k] = \frac{e^{T\Psi(0,-ik)}}{(e^{h_0}-b)^k}\, m_k(T;h_0),$$

where $m_k(T;h_0)$ is computed via the Laplace–resolvent method of Section 5. Consequently, the method computes $\mathbb{E}[\Pi_T^k]$ for every integer

$$k \in \mathscr{K} := \{n \in \mathbb{N} :\ n\eta_{1,+} < 1 \text{ and } n\eta_{2,+} < 1\}.$$

Let $p_{\text{surv}}(h_0) := \mathbb{P}(\tau > T)$ and $p_{\text{liq}}(h_0) := 1 - p_{\text{surv}}(h_0)$. Since Π_T already contains the survival indicator $\mathbf{1}_{\{\tau>T\}}$, the conditional k-th moment on surviving paths is

$$\mathbb{E}[\Pi_T^k \mid \tau > T] = \frac{\mathbb{E}[\Pi_T^k]}{p_{\text{surv}}(h_0)}, \qquad k \geq 1, \quad k\eta_{1,+} < 1, \quad k\eta_{2,+} < 1.$$

In particular,

$$\mathbb{E}[\Pi_T \mid \tau > T] = \frac{\mathbb{E}[\Pi_T]}{p_{\text{surv}}(h_0)}, \qquad \operatorname{Var}(\Pi_T \mid \tau > T) = \frac{\mathbb{E}[\Pi_T^2]}{p_{\text{surv}}(h_0)} - \left(\frac{\mathbb{E}[\Pi_T]}{p_{\text{surv}}(h_0)}\right)^2.$$

The same killed moments also give the unconditional variance of the zero-after-liquidation benchmark,

$$\operatorname{Var}(\Pi_T) = \mathbb{E}[\Pi_T^2] - \mathbb{E}[\Pi_T]^2.$$

Thus survival, killed moments, unconditional moments, and conditional surviving-path moments are all outputs of the same construction. No preference parameter enters this reduction; Section 7 later combines these outputs under several illustrative decision rules.

Remark 4 (No-shorting limit). As $h_0 \to \infty$ the holding map gives $n_2 S_{2,0} \to 0$ and $n_1 S_{1,0} \to 1$, so liquidation becomes impossible and the portfolio converges to a unit long position in asset 1. For every admissible integer $k \geq 1$,

$$\lim_{h_0\to\infty} \mathbb{E}[\Pi_T^k] = \mathbb{E}[e^{kX_{1,T}}] = e^{T\Psi(-ik,0)},$$

which provides a useful diagnostic: the formula for $\mathbb{E}[\Pi_T^k]$ must satisfy this limit exactly. The $k=1$ case gives $e^{\mu_1 T}$ by the drift normalisation (12).

The remainder of the analysis is devoted to computing $m_k(T;h_0)$ for every admissible integer $k \geq 1$ and $p_{\text{surv}}(h_0)$ via the Laplace–resolvent construction.

5. Laplace Transforms of the Killed Moments

This section computes the time-Laplace transform of the reduced killed moment $m_k(T;h_0)$ from Proposition 4.1. We first express that target as a killed resolvent, then evaluate it through a fixed-dimensional linear system. The lower-boundary calculation specific to downward jumps is isolated in Appendix A.

5.1. Target Transform and Resolvent

Fix an integer $k \geq 0$ satisfying (16), and define

$$G_k(x) := (e^x - b)^k = \sum_{j=0}^{k} d_{k,j} e^{jx}, \qquad d_{k,j} := \binom{k}{j} (-b)^{k-j}. \tag{18}$$

For a starting distance $x > 0$ above the translated barrier, define the killed moment function directly by

$$\widetilde{U}_k(t,x) := \mathbb{E}_{\mathbb{P}^{(k)}} \left[G_k(x + Z_t) \mathbf{1}_{\{\inf_{0 \leq s \leq t}(x+Z_s) > 0\}} \right].$$

Because $\sigma_Z > 0$, zero is regular for the negative half-line, so this survival event agrees almost surely with the strict-breach convention $x + Z_s < 0$ used in Section 2. Then $m_k(t;h_0) = \widetilde{U}_k(t,h_0)$ for $k \geq 1$. For uniform notation, set $p_{\text{surv}}(t,h_0) := \mathbb{P}(\tau > t)$ and $m_0(t;h_0) := p_{\text{surv}}(t,h_0)$. Since $G_0 \equiv 1$, the same identity $m_k(t;h_0) = \widetilde{U}_k(t,h_0)$ then holds for every $k \geq 0$ considered below.

For q to the right of the time-Laplace abscissa of convergence, define

$$\widehat{U}_k(q,x) := \int_0^\infty e^{-qt} \widetilde{U}_k(t,x)\,\mathrm{d}t, \qquad \widehat{m}_k(q;h_0) := \int_0^\infty e^{-qt} m_k(t;h_0)\,\mathrm{d}t = \widehat{U}_k(q,h_0). \tag{19}$$

At $k = 0$, (19) is the Laplace transform of the survival probability.

We regard $\widetilde{U}_k(t,\cdot)$ as a function on $\mathbb{R}$ by setting it equal to zero on $(-\infty, 0]$. With killing encoded by this convention,

$$\partial_t \widetilde{U}_k(t,x) = \mathscr{L}^{(k)} \left[\widetilde{U}_k(t,\cdot) \right](x), \qquad x > 0, \tag{20}$$

with initial condition $\widetilde{U}_k(0,x) = G_k(x)$ for $x > 0$. Integrating (20) by parts in t gives

$$q\widehat{U}_k(q,x) - G_k(x) = \mathscr{L}^{(k)} \left[\widehat{U}_k(q,\cdot) \right](x).$$

Equivalently, under the zero-extension convention, the target solves

$$(q - \mathscr{L}^{(k)})\widehat{U}_k(q,x) = G_k(x), \qquad x > 0, \qquad \widehat{U}_k(q,x) = 0 \quad (x \leq 0). \tag{21}$$

Appendix A verifies how the zero extension acts on the jump terms and derives the boundary equations used below.

5.2. Time-Laplace Transform and Finite-Dimensional System

We solve (21) under the following regularity assumption.

Assumption 5.1 (Generic six-root regime)**.** Fix an integer $k \geq 0$ satisfying (16). Let real q lie strictly to the right of the relevant time-Laplace abscissa of convergence. Assume:

(i) $\sigma_Z > 0$;

(ii) the characteristic equation $\psi_Z^{(k)}(\gamma) = q$ has six simple roots $\gamma_1^{(k)}(q), \ldots, \gamma_6^{(k)}(q)$, ordered so that

$$\mathrm{Re}(\gamma_{1,2,3}^{(k)}) > 0, \qquad \mathrm{Re}(\gamma_{4,5,6}^{(k)}) < 0;$$

(iii) $q \neq \psi_Z^{(k)}(j)$ for $j = 0, \ldots, k$.

The forcing expansion (18) and generator eigenrelation (15) motivate the interior ansatz

$$\widehat{U}_k(q,x) = \sum_{j=0}^{k} \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} e^{jx} + \sum_{m=1}^{3} B_m^{(k)}(q) e^{\gamma_{m+3}^{(k)}(q)x}, \qquad x > 0. \tag{22}$$

The exponential-root structure of (22) closely follows the Laplace-transform construction of Sepp (2004) for barrier options under double-exponential jump diffusion, adapted here to a forced resolvent for killed moments. In this adaptation, the first sum is the particular response to the forcing G_k, while the second is a homogeneous barrier correction. As the starting distance $x \to +\infty$, the effect of the lower barrier must vanish and the killed resolvent must approach its unrestricted counterpart. A term $e^{\gamma x}$ with $\mathrm{Re}(\gamma) > 0$ would instead diverge, so the three positive-real-part roots are inadmissible. The ansatz therefore retains only the three roots with negative real part, whose exponentials decay. Their coefficients $B_m^{(k)}(q)$ remain to be determined. The following proposition states the desired transform; Appendix A supplies the downward-phase boundary calculation behind its three matrix rows.

Proposition 5.1 (Laplace-transformed killed moments)**.** *Under Assumption 5.1 and the nondegeneracy condition*

$$r_{1,-}^{(k)} \neq r_{2,-}^{(k)}, \tag{23}$$

the coefficients $\mathbf{B}^{(k)}(q) := (B_1^{(k)}, B_2^{(k)}, B_3^{(k)})^\top$ *are the unique solution of*

$$\mathbf{M}_{\mathrm{bar}}^{(k)}(q)\,\mathbf{B}^{(k)}(q) = -\mathbf{b}^{(k)}(q), \tag{24}$$

where

$$\mathbf{M}_{\mathrm{bar}}^{(k)}(q) = \begin{pmatrix} 1 & 1 & 1 \\ \dfrac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + \gamma_4^{(k)}(q)} & \dfrac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + \gamma_5^{(k)}(q)} & \dfrac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + \gamma_6^{(k)}(q)} \\ \dfrac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + \gamma_4^{(k)}(q)} & \dfrac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + \gamma_5^{(k)}(q)} & \dfrac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + \gamma_6^{(k)}(q)} \end{pmatrix} \tag{25}$$

and

$$\mathbf{b}^{(k)}(q) = \begin{pmatrix} \displaystyle\sum_{j=0}^{k} \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} \\ \displaystyle\sum_{j=0}^{k} \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} \frac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + j} \\ \displaystyle\sum_{j=0}^{k} \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} \frac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + j} \end{pmatrix}. \tag{26}$$

By Lemma 5.2, the matrix (25) is invertible, so

$$\mathbf{B}^{(k)}(q) = -\left(\mathbf{M}_{\mathrm{bar}}^{(k)}(q)\right)^{-1} \mathbf{b}^{(k)}(q).$$

The Laplace-transformed killed moment is therefore

$$\boxed{\widehat{m}_k(q;h_0) = \widehat{U}_k(q,h_0) = \sum_{j=0}^{k} \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} e^{jh_0} + \sum_{m=1}^{3} B_m^{(k)}(q) e^{\gamma_{m+3}^{(k)}(q)h_0}.} \tag{27}$$

▷ *Proof, p. 26*

Lemma 5.2 (Invertibility of $\mathbf{M}^{(k)}_{\text{bar}}(q)$). *Under Assumption 5.1 and condition (23), the barrier matrix $\mathbf{M}^{(k)}_{\text{bar}}(q)$ in (25) is invertible.*

▷ *Proof, p. 27*

The lemma ensures that the absorption condition and the two downward-phase cancellation conditions determine the three decaying-mode coefficients uniquely.

The generic system (24) assumes distinct downward rates. If the two rates coincide, they form one phase and the reduced construction is 2×2; the details are given in Appendix A, Remark 5.

5.3. Moment Recovery and Survival

Numerical Laplace inversion of (27) recovers the time-domain killed moment,

$$m_k(T;h_0) = \mathscr{L}^{-1}_{q\to T}[\widehat{m}_k(q;h_0)]\,. \tag{28}$$

The survival probability is the same construction at $k = 0$. Since $\psi_Z^{(0)}(0) = 0$, one has $P_0(q,x) = 1/q$ and, when $\eta_{1,-} \neq \eta_{2,+}$,

$$\widehat{p}_{\text{surv}}(q;h_0) = \widehat{U}_0(q,h_0) = \frac{1}{q} + \sum_{m=1}^{3} B_m^{(0)}(q)e^{\gamma^{(0)}_{m+3}(q)h_0}, \qquad \mathbf{M}^{(0)}_{\text{bar}}(q)\mathbf{B}^{(0)}(q) = -\begin{pmatrix}1/q\\1/q\\1/q\end{pmatrix}. \tag{29}$$

Hence

$$p_{\text{surv}}(T,h_0) = \mathscr{L}^{-1}_{q\to T}[\widehat{p}_{\text{surv}}(q;h_0)]\,, \qquad p_{\text{liq}}(T,h_0) = 1 - p_{\text{surv}}(T,h_0).$$

If the two untilted downward rates coincide, the $k = 0$ instance of Appendix A, Remark 5, gives the corresponding 2×2 system.

6. Empirical Illustration: Calibration and Health-Buffer Evaluation Map

We illustrate the framework with 540 four-hour WETH/WBTC return pairs and horizon $T = 1/12$. A normalized, causal exponentially weighted mean (EWM) with half-life T produces 539 residual increments by subtracting the lagged EWM mean once from each observed increment. No additional sample demeaning is applied. A penalized empirical characteristic function (ECF) criterion estimates the eleven diffusion, jump, and dependence parameters from those residual increments, following Feuerverger and McDunnough (1981) and Singleton (2001); the residual expected log drifts are fixed at zero. The accompanying empirical-calibration note gives the complete construction.

The EWM step estimates the conditional log-return mean separately from the ECF shape fit. For each asset, the annualized mean retained in this section is

$$g_i := \frac{\mathbb{E}[X_{i,t}]}{t} = d_i^{\mathrm{EWM}} + r_i^{\mathrm{role}}, \qquad \bar{J}_i = p_i \eta_{i,+} - (1 - p_i)\eta_{i,-},$$

where r_i^{role} is the lending rate for the long asset and the borrowing rate for the short asset. Because jumps contribute $\lambda_i \bar{J}_i$ to the expected log return, the finite-variation coefficient used in the log-price SDE is

$$\mu_i^X = g_i - \lambda_i \bar{J}_i.$$

The ECF is fitted only to these EWM residual increments, whose residual expected log-return drift is fixed at zero as a population restriction; it estimates $\sigma_i, \lambda_i, p_i, \eta_{i,+}, \eta_{i,-}$, and ρ, not another directional mean. The implementation converts g_i internally to the expected-price growth convention required by the moment code, but Section 6 reports and perturbs means only in log-return space. We evaluate both role assignments using the role-adjusted $g_1 - g_2$ spread. This selects long WBTC and short WETH. The terminal-block WBTC threshold $b = 0.78$ and maximum LTV 0.73 imply $h_0^{\min} = \log(0.78/0.73) \approx 0.0662$. Table 2 reports the resulting benchmark.

Table 3 compares the semi-analytical (SA) calculation with a discretely monitored Monte Carlo (MC) benchmark at selected points on the plotted health-factor range. The conditional statistics are computed given survival to T.

The penultimate row evaluates the no-shorting proxy $h_0 = 100$ with both methods under the same settings as the other rows. SA agrees with the exact limit to the reported precision; MC differs from the exact mean by $+0.010\%$ and from the exact variance by -0.38%. The final row is the analytical limit: as $h_0 \to \infty$, the portfolio becomes a unit position in WBTC, so its mean is $e^{\mu_1 T} = 1.0125$ and not $e^{g_1 T} = 1.0080$. The latter exponentiates the mean log return; it does not include the diffusion and jump convexity corrections needed for the arithmetic mean of the gross return.

Conditional moments are close. The discretely monitored MC liquidation estimate is lower near the barrier, as expected when crossings between monitoring dates are missed. Runtime is about 0.04 seconds for SA and 13 seconds for MC on this implementation and host.

Group	Parameter	Long S_1: WBTC	Short S_2: WETH
Log-return mean	Endpoint EWM log-return mean d_i^{EWM}	0.095089	−0.555894
	Lending/borrowing rate r_i^{role}	0.0000556	0.027855
	Expected log-return mean g_i	0.095145	−0.528039
	Mean jump rate $\lambda_i \bar{J}_i$	0.926182	0.852120
	Log-SDE coefficient μ_i^X	−0.831037	−1.380159
	Expected-price growth exponent μ_i	0.149275	−0.435038
Kou	Diffusion volatility σ_i	0.256	0.314
	Jump intensity λ_i	321.43	609.87
	Upward-jump probability p_i	0.493	0.413
	Mean upward-jump size $\eta_{i,+}$	0.0106	0.0113
	Mean downward-jump size $\eta_{i,-}$	0.00459	0.00559
Correlation / contract	Brownian correlation ρ	0.842	
	WBTC liquidation threshold b	0.78	
	WBTC maximum origination LTV	0.73	
Portfolio	Initial price $S_{i,0}$	75,060.57	2,060.33
	Horizon T	1/12 (one month)	

Table 2: Empirical benchmark from four-hour WETH/WBTC returns (2026-02-26–2026-05-27). Shape parameters are fitted by penalized ECF to the residual increments obtained after one causal EWM subtraction with half-life T; no additional sample demeaning is imposed. The Log-return mean group keeps the empirical drift construction in log-return space: g_i is the endpoint EWM mean plus the role-specific lending or borrowing rate, and $\mu_i^X = g_i - \lambda_i \bar{J}_i$ is the corresponding finite-variation coefficient in the log-price SDE. The final row gives the distinct price-growth exponent defined by $\mathbb{E}[e^{X_{i,t}}] = e^{\mu_i t}$. The sample-average AAVE rates over the observation window are 0.0056% for WBTC lending (supply) and 2.7855% for WETH variable borrowing. Both are annual APRs expressed as decimal rates in the calculation (year^{-1}), matching the annual time scale of d_i^{EWM} and the Kou parameters; no four-hour scaling is applied before they enter g_i. Contract terms and initial prices are measured at terminal block 25,189,091.
Result-generation script: `jobs/calibrate_eth_btc.py`

h_0	H_0	p_{liq}		Conditional mean		Conditional variance		Time (s)	
		SA	MC	SA	MC	SA	MC	SA	MC
0.0662	1.0685	0.3833	0.3822	1.3072	1.3102	0.0774	0.0781	0.044	13.90
0.0953	1.1000	0.2444	0.2440	1.2372	1.2388	0.0708	0.0715	0.039	13.85
0.1823	1.2000	0.0494	0.0496	1.1263	1.1265	0.0582	0.0591	0.039	13.85
0.4055	1.5000	1.58×10^{-4}	2.40×10^{-4}	1.0647	1.0649	0.0328	0.0330	0.041	14.00
0.6931	2.0000	5.77×10^{-9}	0^{*}	1.0433	1.0434	0.0195	0.0196	0.040	13.91
100	2.69×10^{43}	0	0	1.0125	1.0126	0.0094	0.0093	0.039	14.03
∞	∞	0		1.0125		0.0094		exact limit	

Table 3: Semi-analytical and Monte Carlo calculations for the benchmark in Table 2, with $T = 1/12$. SA uses Talbot inversion with $M = 32$ and times the survival probability and killed moments $k = 1,2$. MC uses 25,000 paths, 1,008 monitoring intervals, and seed 42. Times are single-run wall-clock seconds and are implementation- and machine-dependent. 0^{*} means that no liquidation was observed in the simulated sample, not that the underlying probability is zero. The $h_0 = 100$ row is a finite SA/MC proxy for no shorting; the $h_0 = \infty$ row reports the corresponding analytical limit.

Result-generation script: `jobs/empirical_method_comparison.py`

Applying the resolvent map to this benchmark over $H_0 \in [1.068, 2.000]$ gives the health-buffer evaluation map in Figure 1. Before imposing an investor objective, the three-dimensional curve shows how the initial buffer jointly changes liquidation probability and the conditional mean and variance of the surviving payoff.

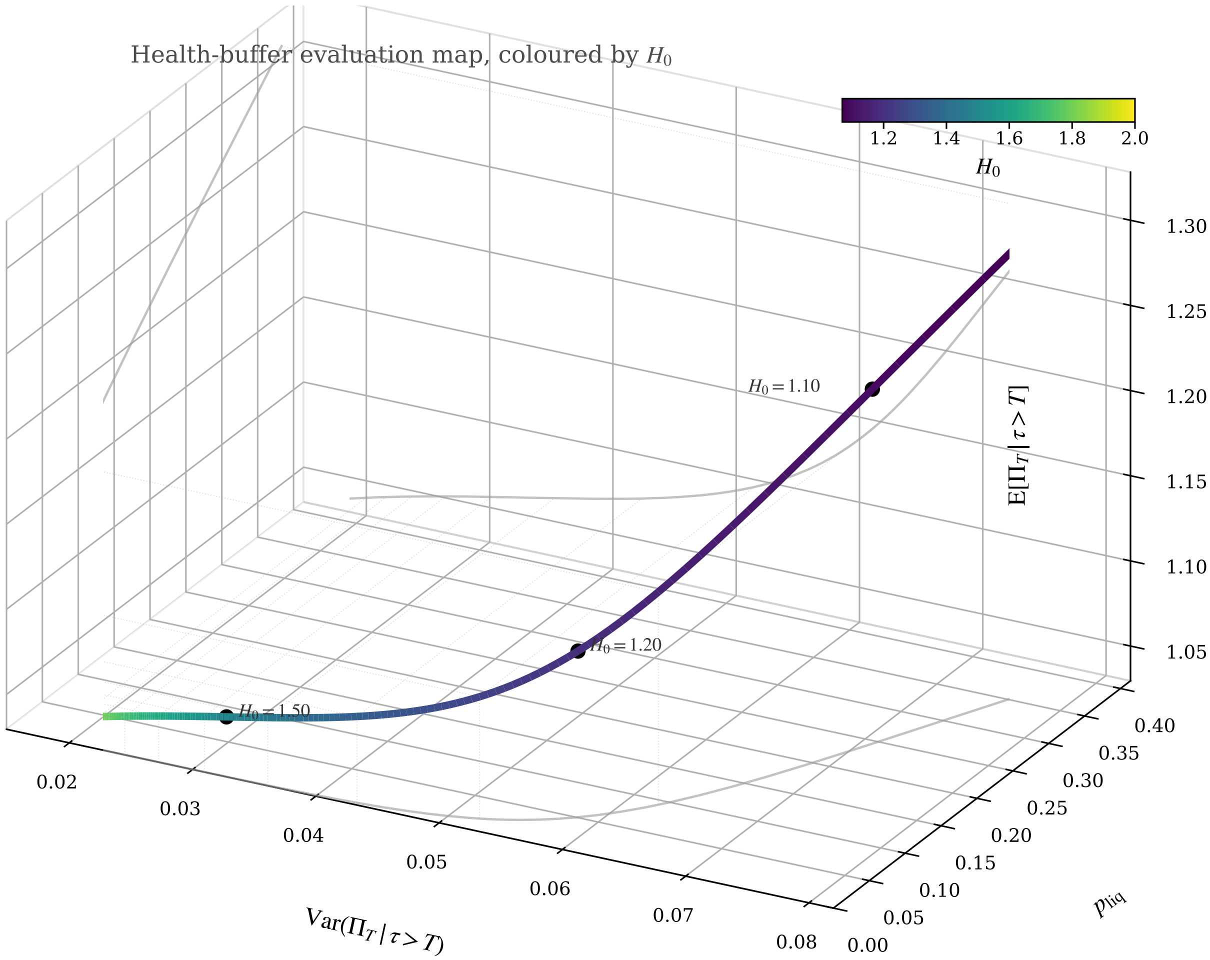

Figure 1: Health-buffer evaluation map for the empirical long-WBTC/short-WETH benchmark in Table 2. Axes: $x = \mathrm{Var}(\Pi_T \mid \tau > T)$, $y = p_{\mathrm{liq}} = 1 - p_{\mathrm{surv}}$, and $z = \mathrm{E}[\Pi_T \mid \tau > T]$. The curve is parametrised and coloured by $H_0 \in [1.068, 2.000]$; labelled points identify representative initial health factors. Grey curves are orthogonal guides on the coordinate planes.

Result-generation script: `jobs/health_buffer_evaluation_map.py`

6.1. Sensitivity to the long–short mean spread

The evaluation map is sensitive to the expected log-return means supplied to the framework. We therefore hold the calibrated mean midpoint and all non-drift inputs fixed and perturb only the long–short mean spread around its calibrated value. The first analysis tracks liquidation risk and conditional payoff moments; the second combines payoff and survival through the expected killed payoff and studies the induced optimal health buffer.

6.1.1 Liquidation risk and conditional moments

To isolate spread uncertainty from a common shift in the log-return mean, let $\Delta_g^{\mathrm{bm}} = g_1^{\mathrm{bm}} - g_2^{\mathrm{bm}} = 0.623184$ and $\bar{g}^{\mathrm{bm}} = (g_1^{\mathrm{bm}} + g_2^{\mathrm{bm}})/2 = -0.216447$. We hold this midpoint and all calibrated diffusion, jump, correlation, contract, and price parameters fixed. Let c denote an additive shock, in annualized log-return units, to the benchmark spread. We set

$$g_1(c) = g_1^{\mathrm{bm}} + \frac{c}{2}, \qquad g_2(c) = g_2^{\mathrm{bm}} - \frac{c}{2}, \qquad \Delta_g(c) = \Delta_g^{\mathrm{bm}} + c, \qquad c \in [-0.70, 0.70]. \tag{30}$$

Thus $c = 0$ is the calibrated benchmark, $c < 0$ narrows the spread, and $c > 0$ widens it; the endpoints apply a symmetric ∓ 35 percentage-point change to the two annualized leg means and move their spread by ∓ 70 percentage points. Hence the scenario range is $\Delta_g(c) \in [-0.076816, 1.323184]$. The spread is zero at $c = -\Delta_g^{\mathrm{bm}} = -0.623184$ and becomes negative below this threshold, reversing the calibrated expected-return ranking. For each c, the implementation maps $g_i(c)$ to its internal price-growth convention while leaving all ECF shape parameters fixed. This is a symmetric fixed-midpoint drift scenario, not a recalibration of those shape parameters; payoff moments can depend separately on both marginal drifts as well as on their difference.

Figure 2 evaluates 57 equally spaced values of $c \in [-0.70, 0.70]$ at $H_0 = 1.10$ and $T = 1/12$. Table 4 reports five representative points. We refer to p_{liq} as default risk in this illustration; operationally, the event is first passage through the protocol liquidation boundary. Narrowing the spread by 0.70 increases p_{liq} from 24.44% to 39.52% and lowers the conditional mean from 1.2372 to 1.1313. Widening it by 0.70 lowers p_{liq} to 13.85% and raises the conditional mean to 1.3632. Relative to the benchmark, these endpoints change liquidation probability by +61.7% and −43.3%, respectively. Conditional variance changes by approximately −17.6% and +17.7%; conditional skewness and excess kurtosis also move appreciably. Mean uncertainty therefore propagates not only to default probability and expected surviving payoff, but to the dispersion and shape of the conditional payoff distribution.

c	$g_1 - g_2$	p_{liq}	$\mathbb{E}[\Pi_T \mid \tau > T]$	$\mathrm{Var}(\Pi_T \mid \tau > T)$	Skew.	Ex. kurt.
−0.70	−0.0768	0.3952	1.1313	0.0584	0.7226	0.6766
−0.35	0.2732	0.3147	1.1817	0.0645	0.6640	0.5576
0.0	0.6232	0.2444	1.2372	0.0708	0.6052	0.4572
0.35	0.9732	0.1857	1.2978	0.0771	0.5482	0.3784
0.70	1.3232	0.1385	1.3632	0.0833	0.4950	0.3222

Table 4: Selected expected-log-return-mean spread sensitivity results. The calibrated case is $c = 0$; all figures use the same semi-analytical inversion and fixed non-drift inputs as the benchmark.

Result-generation script: `jobs/mu_ spread_ sensitivity.py`

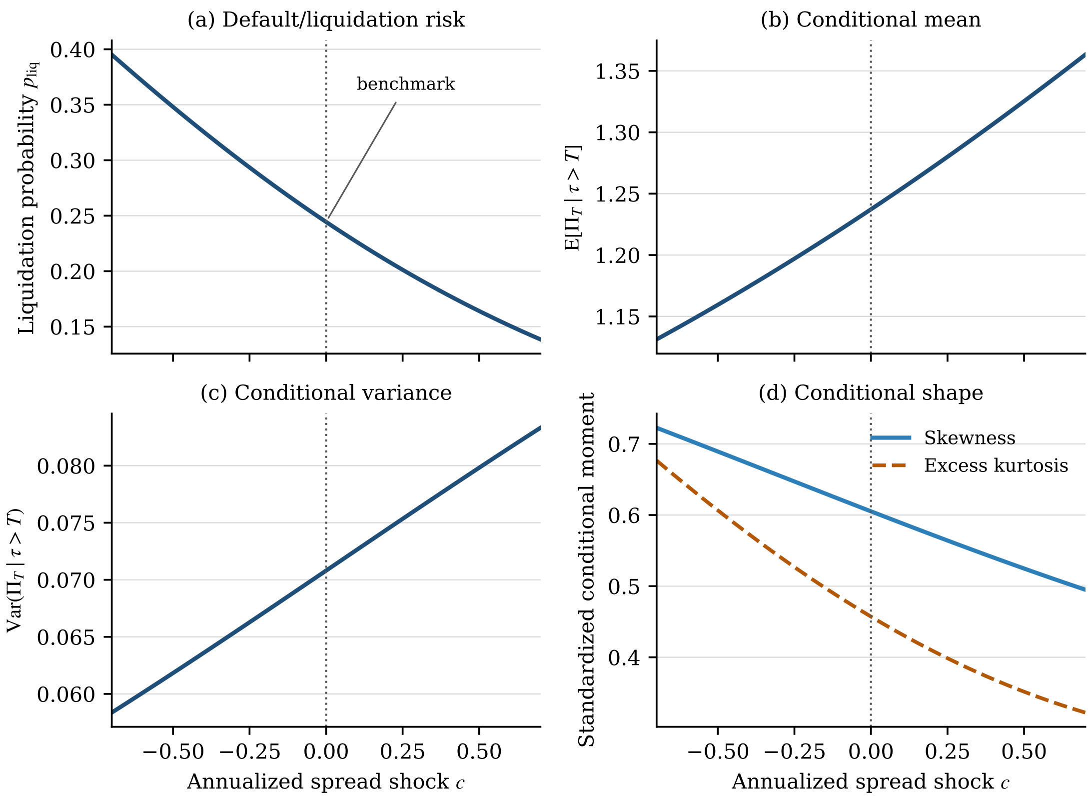

Figure 2: Sensitivity to the supplied expected log-return mean spread at $H_0 = 1.10$ and $T = 1/12$. The midpoint of (g_1, g_2) and every non-drift input are held at their calibrated values while the benchmark spread receives the additive annualized shock c as in (30). The dotted vertical line marks the calibrated benchmark $c = 0$. Conditional moments are computed given $\tau > T$.

Result-generation script: `jobs/mu_spread_sensitivity.py`

6.1.2 Expected killed payoff and optimal health buffer

Let $m(c,h_0) := \mathbb{E}[\Pi_T \mid \tau > T]$ and $p_{\text{surv}}(c,h_0) := 1 - p_{\text{liq}}(c,h_0)$. We study the sensitivity of the optimal initial log health buffer to the first killed moment

$$M_1(T,h_0) := p_{\text{surv}}(c,h_0)m(c,h_0) = \mathbb{E}[\Pi_T] =: J_{\text{kill}}(c,h_0). \tag{31}$$

If the terminal portfolio payoff is regarded as a contingent claim, M_1 has the payoff structure of a zero-rebate barrier option: crossing the liquidation boundary knocks out the claim and sets its terminal payoff to zero. Because the expectation is taken under the physical measure, M_1 is an expected-performance criterion rather than an arbitrage-free option price.

At fixed $H_0 = 1.10$, Figure 3 uses the same 57-point spread grid, $T = 1/12$, and calibrated non-drift inputs as Section 6.1.1. The calibrated benchmark gives $J_{\text{kill}} = 0.9349$. A -0.70 spread shock lowers it to 0.6843, or 26.8% below the benchmark, while a $+0.70$ shock raises it to 1.1744, or 25.6% above the benchmark.

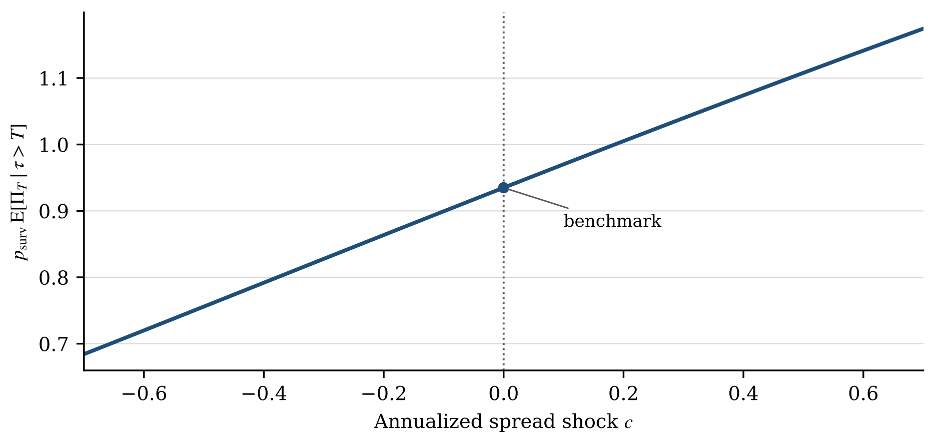

Figure 3: Expected killed payoff at $H_0 = 1.10$ as the expected-log-return mean spread varies. The calibrated benchmark is $c = 0$; negative values narrow and positive values widen the calibrated spread. All non-drift inputs are fixed, with $T = 1/12$.

Result-generation script: `jobs/mu_spread_sensitivity.py`

To examine the induced buffer choice, define

$$h_0^\star(c) \in \arg\max_{h_0 \in [h_0^{\min}, \log 2]} J_{\text{kill}}(c,h_0), \qquad H_0^\star(c) := e^{h_0^\star(c)}. \tag{32}$$

We solve (32) by bounded scalar optimization over $H_0 \in [1.0685, 2]$, with absolute tolerance 10^{-7} and explicit checks of both endpoints. At the calibrated benchmark, the solution is $H_0^\star = 1.2711$, $h_0^\star = 0.2399$, initial leverage 2.5882, and $J_{\text{kill}}^\star = 1.0811$. The optimizer is spread-sensitive: representative $H_0^\star$ values are 2.0000, 1.3624, 1.2711, 1.2149, and 1.1751 at $c = -0.70, -0.35, 0, 0.35, 0.70$, respectively. Hence a wider expected-log-return spread supports more initial leverage and a smaller health buffer, while a narrower spread favors deleveraging. The corresponding maximized killed payoff rises from 0.9773 at $c = -0.70$ to 1.2278 at $c = 0.70$. On the evaluated grid the upper health-factor constraint binds for $c \le -0.60$; the other four representative optimizers are interior.

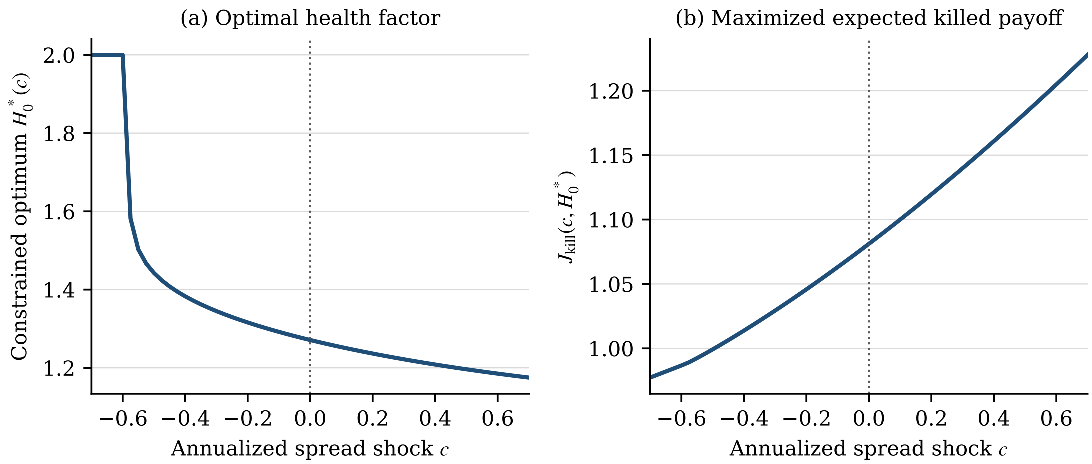

Figure 4: Sensitivity of the expected-killed-payoff optimizer to the expected-log-return mean spread. Panel (a) reports the bounded continuous solution $H_0^*(c)$; panel (b) reports the maximized expected killed payoff. The dotted line marks the calibrated benchmark $c = 0$.

Result-generation script: `jobs/mu_spread_sensitivity.py`

7. Application: Initial Health-Buffer Sizing

Using the empirical benchmark and evaluation map in Section 6, we now give three stylized initial-buffer rules. They illustrate how the common evaluation map supports different preferences, not competing claims about a uniquely correct objective.

7.1. Drift construction and interpretation

The directional drift is deliberately kept outside the ECF optimization. High-frequency sampling sharpens volatility estimates much more than expected-return estimates (Merton, 1980); in a short sample, allowing the characteristic-function phase to select a free drift can therefore confound location with jump-shape discrepancies. Let d_i^{EWM} denote the annualized endpoint EWM log-return signal and $\bar{J}_i = p_i \eta_{i,+} - (1-p_i)\eta_{i,-}$. Because financing is role-dependent, the expected log-return means are constructed for each complete ordered assignment (L,S):

$$g_L(L,S) = d_L^{\text{EWM}} + r_{\text{supply},L}, \qquad g_S(L,S) = d_S^{\text{EWM}} + r_{\text{borrow},S}.$$

Writing $\mathscr{O} = \{(\text{WETH}, \text{WBTC}), (\text{WBTC}, \text{WETH})\}$, the showcase selects

$$(L^\star, S^\star) = \arg\max_{(L,S)\in\mathscr{O}} \{g_L(L,S) - g_S(L,S)\},$$

provided that the maximum is positive. After selection, the expected-price growth exponent required internally by the moment code is derived, rather than estimated, as

$$\mu_i = g_i + \mu_i^{\text{res}}, \qquad \mu_i^{\text{res}} = \frac{1}{2}\sigma_i^2 + \lambda_i(\chi_i - \bar{J}_i),$$

and the log-SDE coefficient is $\mu_i^X = g_i - \lambda_i \bar{J}_i$. Thus the orientation rule and its reported mean remain in log-return space, while the price-growth conversion is only an implementation step.

Exponential-smoothing choices generally depend on the forecast horizon (Cox, 1961; Tiao and Xu, 1993); the half-life-T specification used here is a transparent horizon-matching convention, not an estimated optimum. The 90-day sample contains only about three non-overlapping one-month forecast horizons, too few for a persuasive horizon-specific out-of-sample selection. A practical implementation should therefore compare half-lives $T/2$, T, and $2T$, as well as a zero-directional-trend, carry-only case. The reported benchmark is one empirical scenario rather than a claim that its g_i are structural or precisely estimated expected returns.

Write

$$\mu_c(h_0) := \mathbb{E}[\Pi_T \mid \tau > T], \qquad v_c(h_0) := \mathrm{Var}(\Pi_T \mid \tau > T),$$

and, under the killed-payoff convention,

$$\mu_u(h_0) := M_1(T,h_0), \qquad v_u(h_0) := M_2(T,h_0) - M_1(T,h_0)^2.$$

The subscripts distinguish moments conditional on survival from unconditional moments of the random variable that is set to zero after liquidation.

7.2. Candidate decision rules

Liquidation-constrained rule. Given a surviving-path score J_{surv} and a liquidation budget $\bar{p}$, select an objective-specific buffer from

$$\operatorname*{arg\,max}_{h_0 \in \mathscr{H}(b,\ell_{\max})} \; J_{\mathrm{surv}}(h_0) \quad \text{subject to} \quad p_{\mathrm{liq}}(T,h_0) \le \bar{p}. \tag{33}$$

For example, a policy that seeks the greatest initial exposure allowed by a risk limit sets $J_{\mathrm{surv}} = L_0$.

Unconditional killed-payoff mean–variance rule. For $\alpha \ge 0$, select from

$$\operatorname*{arg\,max}_{h_0 \in \mathscr{H}(b,\ell_{\max})} \left\{ \mu_u(h_0) - \frac{\alpha}{2} v_u(h_0) \right\}. \tag{34}$$

This rule treats the killed payoff as terminal wealth. Its interpretation therefore depends on the deliberately severe zero-after-liquidation convention and need not coincide with a criterion based on a protocol recovery model.

Conditional performance with an explicit liquidation penalty. For $\alpha, \delta \ge 0$, select from

$$\operatorname*{arg\,max}_{h_0 \in \mathscr{H}(b,\ell_{\max})} \left\{ \mu_c(h_0) - \frac{\alpha}{2} v_c(h_0) - \delta\, p_{\mathrm{liq}}(T,h_0) \right\}. \tag{35}$$

Here δ is an explicit preference parameter, not a quantity identified by the price model. Conditional moments can also display survivor selection near the barrier; the penalty controls how the decision rule weighs that feature against liquidation incidence. The paper does not claim that any one of (33)–(35) is universally correct. Their purpose is to show how the same semi-analytic outputs enter different ex-ante preferences.

7.3. Illustrative rule-specific selections

Table 5 applies the preceding rules to the empirical benchmark in Table 2 and the same buffer grid used for Figure 1. A value selected at a grid endpoint is reported as such; it should not be read as an interior or universal optimum.

Rule	Parameters	Selected H_0	Initial L_0	p_{liq}
Maximum leverage under risk limit	$\bar{p} = 0.05$	1.200	2.859	0.0498
Maximum leverage under risk limit	$\bar{p} = 0.10$	1.162	3.041	0.0931
Maximum leverage under risk limit	$\bar{p} = 0.20$	1.115	3.326	0.1945
Unconditional mean–variance	$\alpha = 0.5$	1.312	2.466	0.0065
Unconditional mean–variance	$\alpha = 1$	1.345	2.381	0.0035
Unconditional mean–variance	$\alpha = 5$	2.000	1.639	$< 10^{-4}$
Conditional mean–variance–penalty	$(\alpha, \delta) = (1, 0)$	1.068	3.704	0.3833
Conditional mean–variance–penalty	$(\alpha, \delta) = (20, 0)$	2.000	1.639	$< 10^{-4}$
Conditional mean–variance–penalty	$(\alpha, \delta) = (1, 0.25)$	1.068	3.704	0.3833
Conditional mean–variance–penalty	$(\alpha, \delta) = (1, 0.5)$	1.111	3.359	0.2087

Table 5: Objective-specific selections for the empirical long-WBTC/short-WETH benchmark over the 200-point grid $H_0 \in [1.068, 2.000]$. Only the decision rule or its stated parameters change. Endpoint selections underscore that this is an illustration rather than a preferred practical buffer; the constrained rows maximize L_0 subject to the stated liquidation budget.

Result-generation script: `jobs/objective_comparison.py`

8. Conclusion and Discussion

We derive an exact Laplace-domain representation and a semi-analytic numerical procedure for survival probabilities and killed payoff moments of a leveraged crypto long–short position whose liquidation boundary is determined by a collateral-to-liability ratio. The initial log health factor reduces the holdings to one parameter, and an exponential measure change reduces each fixed admissible integer-order moment to a killed expectation of the log-price ratio. Under Kou jump–diffusion dynamics and distinct downward phase rates, three barrier conditions give a 3×3 resolvent system.

These objective-independent quantities turn initial-buffer sizing into a one-dimensional optimization: rapid evaluation of survival and killed-moment outputs supports buffer selection under alternative liquidation constraints and moment-based preferences. For the empirical long-WBTC/short-WETH benchmark, the health-buffer evaluation map displays the trade-off between the initial health buffer, leverage, liquidation probability, and surviving-path payoff moments. The selected buffer moves with the objective, its risk parameters, and the loss convention; no single selection should be read as a universal practical optimum.

The model omits liquidation recovery, dynamic collateral intervention, oracle and keeper delays, stochastic borrowing rates, execution costs, and market impact. Incorporating these features would change investor terminal wealth and may change the selected buffer. The principal output is therefore not a single leverage number, but the map from the initial health buffer to survival probability and payoff-distribution characteristics – the input that any sizing rule, whether the illustrative ones used here or a more complete dynamic control model, ultimately needs to solve for the optimal buffer.

Appendix A: Killing at the Barrier and Downward-Phase Corrections

Section 5 uses only the generator $\mathscr{L}^{(k)}$ from Lemma 3.1, with a killed expectation extended by zero below the barrier. This appendix records how that convention changes the downward jump integrals and derives the three rows of the barrier system (24).

A.1. Phase Operators and Zero Extension

Definition A.1 (Phase operators)**.** For $j = 1, 2$, the phase operators associated with the tilted jump kernels of Z under $\mathbb{P}^{(k)}$ are

$$K_{j,+}^{(k)}[f](x) := \int_0^\infty f(x+y) r_{j,+}^{(k)} e^{-r_{j,+}^{(k)} y}\,\mathrm{d}y, \qquad K_{j,-}^{(k)}[f](x) := \int_0^\infty f(x-y) r_{j,-}^{(k)} e^{-r_{j,-}^{(k)} y}\,\mathrm{d}y.$$

For any complex γ with $\mathrm{Re}(\gamma) < r_{j,+}^{(k)}$ and $\mathrm{Re}(\gamma) > -r_{j,-}^{(k)}$, direct integration gives

$$K_{j,+}^{(k)}[e^{\gamma\cdot}](x) = \frac{r_{j,+}^{(k)}}{r_{j,+}^{(k)} - \gamma} e^{\gamma x}, \qquad K_{j,-}^{(k)}[e^{\gamma\cdot}](x) = \frac{r_{j,-}^{(k)}}{r_{j,-}^{(k)} + \gamma} e^{\gamma x}. \tag{36}$$

This phase construction originates in Kou and Wang (2003) for first-passage problems under the double-exponential model and is used by Sepp (2004) to reduce barrier pricing to a finite linear system.

The tilted phase intensities are

$$\lambda_{1,+}^{(k)} = \lambda_1 p_1, \quad \lambda_{1,-}^{(k)} = \lambda_1(1-p_1), \quad \lambda_{2,+}^{(k)} = \frac{\lambda_2(1-p_2)}{1+k\eta_{2,-}}, \quad \lambda_{2,-}^{(k)} = \frac{\lambda_2 p_2}{1-k\eta_{2,+}}. \tag{37}$$

Using Definition A.1, the full-line action of the generator is

$$\mathscr{L}^{(k)} f(x) = \mu_Z^{(k)} f'(x) + \frac{\sigma_Z^2}{2} f''(x) + \sum_{a=1}^{2} \lambda_{a,+}^{(k)} \big(K_{a,+}^{(k)}[f](x) - f(x)\big) + \sum_{d=1}^{2} \lambda_{d,-}^{(k)} \big(K_{d,-}^{(k)}[f](x) - f(x)\big). \tag{38}$$

Applying (36) recovers the exponential symbol (15).

Now let f be defined on the alive half-line $x > 0$ and let $\bar{f} = f\mathbf{1}_{(0,\infty)}$ be its zero extension. Upward jumps remain in the alive region, whereas a downward jump of size $y \geq x$ lands in the absorbing region. Consequently,

$$K_{d,-}^{(k)}[\bar{f}](x) = K_{d,-}^{(k),D}[f](x) := \int_0^x f(x-y) r_{d,-}^{(k)} e^{-r_{d,-}^{(k)} y}\,\mathrm{d}y, \qquad x > 0. \tag{39}$$

Substitution into (38) gives

$$\begin{aligned} \mathscr{L}^{(k)}[\bar{f}](x) &= \mu_Z^{(k)} f'(x) + \frac{\sigma_Z^2}{2} f''(x) + \sum_{a=1}^{2} \lambda_{a,+}^{(k)} \big(K_{a,+}^{(k)}[f](x) - f(x)\big) \\ &\quad + \sum_{d=1}^{2} \lambda_{d,-}^{(k)} \big(K_{d,-}^{(k),D}[f](x) - f(x)\big), \qquad x > 0. \end{aligned} \tag{40}$$

Thus killing changes the arguments on which $\mathscr{L}^{(k)}$ acts, not the identity of the generator. Equation (40) is the standard zero-extension construction for a Lévy process killed at an absorbing boundary.

A.2. Downward-Phase Residual and Barrier Conditions

Lemma A.1 (Truncated downward phase)**.** *For $r > 0$, $\beta \in \mathbb{C} \setminus \{-r\}$, and $x > 0$,*

$$\int_0^x e^{\beta(x-y)} r e^{-ry}\,\mathrm{d}y = \frac{r}{r+\beta} e^{\beta x} - \frac{r}{r+\beta} e^{-rx}. \tag{41}$$

No restriction $\mathrm{Re}(\beta) > -r$ *is needed, since the integral is over a finite interval.*

▷ *Proof, p. 26*

For a finite exponential sum $v(x) = \sum_{\beta\in\mathscr{E}} a_\beta e^{\beta x}$, define the truncation-correction coefficient

$$\mathfrak{C}_r[v] := \sum_{\beta\in\mathscr{E}} a_\beta \frac{r}{r+\beta}. \tag{42}$$

Then (41) gives

$$K^D_{r,-}[v](x) := \int_0^x v(x-y) r e^{-ry}\,\mathrm{d}y = \sum_{\beta\in\mathscr{E}} a_\beta \frac{r}{r+\beta} e^{\beta x} - \mathfrak{C}_r[v] e^{-rx}. \tag{43}$$

Combining (40) and (43), for exponents at which the upward integrals are finite and which avoid the downward phase poles,

$$\mathscr{L}^{(k)}[\bar{v}](x) = \sum_{\beta\in\mathscr{E}} a_\beta \psi_Z^{(k)}(\beta) e^{\beta x} - \sum_{d=1}^{2} \lambda_{d,-}^{(k)} \mathfrak{C}_{r_{d,-}^{(k)}}[v] e^{-r_{d,-}^{(k)} x}, \qquad x > 0, \tag{44}$$

where $\psi_Z^{(k)}$ is understood through the meromorphic continuation of its exponential symbol outside the common full-line moment strip.

Apply (44) term by term to the interior ansatz (22). Its forcing coefficients and characteristic-root equations give

$$(q - \mathscr{L}^{(k)})\widehat{U}_k(q,x) = G_k(x) + \sum_{d=1}^{2} \lambda_{d,-}^{(k)} \mathfrak{C}_{r_{d,-}^{(k)}}[\widehat{U}_k(q,\cdot)] e^{-r_{d,-}^{(k)} x}. \tag{45}$$

When the two downward rates are distinct, the residual exponentials are linearly independent, so the interior equation holds if and only if both correction coefficients vanish. Since $\sigma_Z > 0$, the lower boundary is regular for the Brownian component and absorption also imposes the Dirichlet condition. The three conditions are therefore

$$\widehat{U}_k(q,0) = 0, \qquad \mathfrak{C}_{r_{1,-}^{(k)}}[\widehat{U}_k(q,\cdot)] = 0, \qquad \mathfrak{C}_{r_{2,-}^{(k)}}[\widehat{U}_k(q,\cdot)] = 0. \tag{46}$$

Substituting (22) into (46) gives

$$\sum_{m=1}^{3} B_m^{(k)} = -\sum_{j=0}^{k} \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)},$$

and, for $d = 1, 2$,

$$\sum_{m=1}^{3} \frac{r_{d,-}^{(k)}}{r_{d,-}^{(k)} + \gamma_{m+3}^{(k)}(q)} B_m^{(k)} = -\sum_{j=0}^{k} \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} \frac{r_{d,-}^{(k)}}{r_{d,-}^{(k)} + j}.$$

These are precisely the three rows of (24).

Remark 5 (Coincident downward rates). If $r_{1,-}^{(k)} = r_{2,-}^{(k)} =: r$, the two downward kernels combine into a single phase. The meromorphic symbol must first be reduced by cancelling the repeated phase factor; otherwise clearing denominators introduces a spurious root at the pole $-r$. The reduced characteristic equation has two genuine decaying roots, say $\gamma_4^{(k)}(q), \gamma_5^{(k)}(q)$, and the conditions $\widehat{U}_k(q,0) = 0$ and $\mathfrak{C}_r[\widehat{U}_k(q,\cdot)] = 0$ give the 2×2 system

$$\begin{pmatrix} 1 & 1 \\ \dfrac{r}{r+\gamma_4^{(k)}(q)} & \dfrac{r}{r+\gamma_5^{(k)}(q)} \end{pmatrix} \begin{pmatrix} B_1^{(k)} \\ B_2^{(k)} \end{pmatrix} = - \begin{pmatrix} \displaystyle\sum_{j=0}^{k} \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} \\ \displaystyle\sum_{j=0}^{k} \frac{d_{k,j}}{q - \psi_Z^{(k)}(j)} \frac{r}{r+j} \end{pmatrix}.$$

Thus the 3×3 description applies to the generic distinct-rate regime; the coincident-rate case is a genuine lower-dimensional phase model, not a singular generic system.

Appendix B: Proofs

Proof of Lemma 3.1. Step 1: Laplace transform under $\mathbb{P}^{(k)}$. Because $Z_t = X_{1,t} - X_{2,t}$ and the density on $\mathscr{F}_t$ equals $e^{kX_{2,t} - t\Psi(0,-ik)}$,

$$\mathbb{E}_{\mathbb{P}^{(k)}}\left[e^{sZ_t}\right] = e^{-t\Psi(0,-ik)}\,\mathbb{E}_{\mathbb{P}}\left[e^{sX_{1,t}+(k-s)X_{2,t}}\right] = e^{t[\Psi(-is,-i(k-s))-\Psi(0,-ik)]}, \tag{47}$$

where we used $\mathbb{E}_{\mathbb{P}}[e^{\alpha X_{1,t}+\beta X_{2,t}}] = e^{t\Psi(-i\alpha,-i\beta)}$. This identity is first valid for s in the common moment-generating strip. In particular, $k\eta_{2,+} < 1$ is needed at $s = 0$ to define the tilt, and evaluating the payoff forcing at $s = j$, $j = 0,\dots,k$, additionally requires $k\eta_{1,+} < 1$. Since the Radon–Nikodým density is exponential-linear in $(X_{1,t}, X_{2,t})$, Girsanov's theorem for Lévy processes guarantees that Z remains a Lévy process under $\mathbb{P}^{(k)}$ with exponent

$$\psi_Z^{(k)}(s) = \Psi(-is, -i(k-s)) - \Psi(0,-ik). \tag{48}$$

Step 2: Expand and simplify. Substituting (13) with $(u,v) = (-is, -i(k-s))$, and noting $u^2 = -s^2$, $v^2 = -(k-s)^2$, $uv = -s(k-s)$, and $\varphi_{J_i}(-i\cdot) = M_i(\cdot)$:

$$\begin{aligned}\Psi(-is,-i(k-s)) = \mu_1^X s + \mu_2^X (k-s) + \tfrac{1}{2}\sigma_1^2 s^2 + \tfrac{1}{2}\sigma_2^2 (k-s)^2 + \rho\sigma_1\sigma_2 s(k-s)\\ + \lambda_1(M_1(s)-1) + \lambda_2(M_2(k-s)-1).\end{aligned} \tag{49}$$

Setting $s = 0$ in (49) gives $\Psi(0,-ik) = \mu_2^X k + \frac{1}{2}\sigma_2^2 k^2 + \lambda_2(M_2(k)-1)$. Subtracting and collecting terms by type:

Drift. $\mu_1^X s + \mu_2^X (k-s) - \mu_2^X k = (\mu_1^X - \mu_2^X)s.$

Diffusion. Expanding $(k-s)^2 = k^2 - 2ks + s^2$ and $(k-s) = k - s$:

$$\begin{aligned}&\tfrac{1}{2}\sigma_1^2 s^2 + \tfrac{1}{2}\sigma_2^2 (k-s)^2 + \rho\sigma_1\sigma_2 s(k-s) - \tfrac{1}{2}\sigma_2^2 k^2\\ &\quad = \tfrac{1}{2}(\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2)s^2 - k(\sigma_2^2 - \rho\sigma_1\sigma_2)s = \tfrac{1}{2}\sigma_Z^2 s^2 - k(\sigma_2^2 - \rho\sigma_1\sigma_2)s.\end{aligned}$$

Jumps. $\lambda_1(M_1(s)-1) + \lambda_2(M_2(k-s)-1) - \lambda_2(M_2(k)-1) = \lambda_1(M_1(s)-1) + \lambda_2(M_2(k-s) - M_2(k)).$

Combining, the drift coefficient is $\mu_Z^{(k)} := (\mu_1^X - \mu_2^X) - k(\sigma_2^2 - \rho\sigma_1\sigma_2)$, which establishes (14). The eigenrelation (15) follows because for any Lévy process with exponent ψ, the generator satisfies $\mathscr{L}e^{sz} = \psi(s)\,e^{sz}$, as can be seen by differentiating $\mathbb{E}[e^{sZ_t}] = e^{t\psi(s)}$ with respect to t at $t = 0$. □

Proof of Lemma A.1. Direct integration over the finite interval gives

$$\int_0^x e^{\beta(x-y)} r e^{-ry}\,\mathrm{d}y = e^{\beta x} r \int_0^x e^{-(r+\beta)y}\,\mathrm{d}y = e^{\beta x}\frac{r}{r+\beta}\left(1 - e^{-(r+\beta)x}\right),$$

which is (41). Since the interval is finite, this calculation is valid for every $\beta \neq -r$ without a full-line integrability condition. □

Proof of Proposition 5.1. For the ansatz (22), the zero-extension identity (45) shows that the interior resolvent equation holds precisely when its two downward-phase correction coefficients vanish. Absorption adds $\widehat{U}_k(q,0) = 0$, giving the three conditions (46). As shown in Appendix A, substitution of the ansatz produces exactly the system (24) with matrix (25) and forcing vector (26). Lemma 5.2 gives the unique coefficient vector.

The resulting zero extension solves (21); uniqueness of the killed resolvent in the relevant exponential-growth class identifies it with $\widehat{U}_k$. Finally, (19) and evaluation at $x = h_0$ give (27). □

Proof of Lemma 5.2. For brevity, write $\gamma_m := \gamma_m^{(k)}(q)$. Subtracting row 1 from rows 2 and 3 replaces entry (i,m) for $i = 2,3$ by

$$-\frac{\gamma_{m+3}}{r_{i-1,-}^{(k)} + \gamma_{m+3}}.$$

Pulling out -1 from each of these two rows (net sign $(-1)^2 = 1$) and then factoring γ_{m+3} from each column gives

$$\det\big(\mathbf{M}_{\mathrm{bar}}^{(k)}\big) = \gamma_4\gamma_5\gamma_6 \det\left(\frac{1}{x_i + \gamma_{j+3}}\right)_{1\le i,j\le 3},$$

where $x_1 := 0$, $x_2 := r_{1,-}^{(k)}$, $x_3 := r_{2,-}^{(k)}$. The right-hand factor is a Cauchy determinant; the Cauchy formula yields

$$\det\left(\frac{1}{x_i + \gamma_{j+3}}\right) = \frac{\prod_{1\le i<i'\le 3}(x_{i'} - x_i) \cdot \prod_{4\le m<m'\le 6}(\gamma_{m'} - \gamma_m)}{\prod_{i=1}^{3}\prod_{m=4}^{6}(x_i + \gamma_m)}.$$

Since $x_1 = 0$, the factor $\prod_{m=4}^{6}(x_1 + \gamma_m) = \gamma_4\gamma_5\gamma_6$ cancels with the prefactor, yielding a ratio whose every factor is nonzero: $r_{1,-}^{(k)} > 0$ since $\eta_{1,-} > 0$; $r_{2,-}^{(k)} > 0$ since $k\eta_{2,+} < 1$; $(r_{2,-}^{(k)} - r_{1,-}^{(k)}) \neq 0$ by (23); the γ_m are pairwise distinct by Assumption 5.1(ii); and $r_{j,-}^{(k)} + \gamma_m \neq 0$ because $-r_{j,-}^{(k)}$ is a pole of $\psi_Z^{(k)}$, so $\psi_Z^{(k)}(-r_{j,-}^{(k)}) = \infty \neq q$, hence $\gamma_m \neq -r_{j,-}^{(k)}$. □

The survival probability requires no separate proof: it is the $k = 0$ instance of Proposition 5.1 and Lemma 5.2 above. At $k = 0$, the forcing term in (22) is $1/q$ because $\psi_Z^{(0)}(0) = 0$, while (23) becomes $\eta_{1,-} \neq \eta_{2,+}$.

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